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Analysing Tree Cover Changes in Relation to Livestock Grazing Regimes in New South Wales

Bachelor Thesis

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ABSTRACT

Scattered trees are keystone structures because of their disproportionate effect on the ecosystem. They are predicted to decline rapidly due to death of mature trees and lack of regeneration driven partly by agricultural intensification. Rotational grazing and holistic management could possibly promote tree regeneration but are highly contested because there is no consistent empirical evidence. Following up on Fischer et al. (2009), I analysed remotely-sensed tree cover changes from 2005 to 2023 in the Upper Lachlan Catchment in New South Wales, Australia, and their relation to tree density, rotational grazing and holistic management. First, I found that the tree cover may have increased in the study area. Second, there was no significant decline in tree cover with more annual grazing days. That means the positive effect of rotational grazing on tree regeneration found by Fischer et al. (2009) was not visible in the tree cover within the observed time period. Third, the relationship was not more pronounced on sites with scattered trees or grazed woodland. Fourth, tree cover increased slightly on holistic management and slow-rotation farms, whereas it decreased on continuous farms. This might be due to the systems thinking and adaptive management of holistic managers. Avenues for future research may be further monitoring of tree cover change and the analysis of effects of adaptive management, including its effect on tree regeneration. Although the evidence is inconclusive, for now, the best chance of conserving scattered trees might lie in the adoption of holistic management including fast-rotational grazing, reduced fertilization and adaptive decision-making.

1 INTRODUCTION

Scattered tree ecosystems occur in natural and human-modified landscapes throughout the world (Manning et al., 2006). Their common feature is their dispersed pattern of trees (Manning et al., 2006). Synonyms for scattered trees include isolated trees, pasture trees, paddock trees and remnant trees (Manning et al., 2006).

Scattered trees are keystone structures due to their disproportionate effect on ecosystem functions and services at multiple scales and through time. Ecosystem functions and services at the local scale include a wetter microclimate, nutrient enrichment, carbon sequestration, stock shade and habitat provision for animals (England et al., 2020; Manning et al., 2006). For example, scattered trees maintained moderate levels of bird and bat activity in an Australian livestock grazing landscape (Fischer, Stott, et al., 2010). A recent meta-analysis showed that areas with scattered trees support a higher diversity of vertebrates, arthropods and plants than open areas (Prevedello et al., 2018). On the landscape scale, scattered trees contribute to the total tree cover (Manning et al., 2006). For instance, they account for approximately 30% of the tree cover in the Upper Lachlan catchment in New South Wales (Fischer, Sherren, et al., 2010). Furthermore, scattered trees increase landscape connectivity and provide ecological continuity through time by providing refuges for woodland organisms (Manning et al., 2006, 2009). Due to their disproportionate effect on the ecosystem relative to the small area occupied Manning et al. (2006) argued that scattered trees are keystone structures.

Scattered trees are predicted to decline rapidly around the world with severe consequences. Gibbons et al. (2008) simulated rates of tree decline in agricultural landscapes in Spain, the United States, Australia and Costa Rica. They predicted that mature trees will be lost from these landscapes in 90-180 years under current management (Gibbons et al., 2008). Even if their proposed management strategies are implemented, numbers of mature trees will decline before they increase, creating a bottleneck (Gibbons et al., 2008). A loss of these keystone structures implies a reduction in the associated ecosystem functions and tree-dependent species. For example, trees on an average Australian farm will get reduced to two-thirds of their present number which will lead to a decline in birds and bats by up to 50% (Fischer, Zerger, et al., 2010).

The decline of scattered trees is associated with the death of mature trees and a lack of regeneration driven partly by agricultural intensification. Scattered trees are cleared by humans, die due to poor tree health or due to old age (Gibbons et al., 2008; Manning et al., 2006). At the same time, the lack of natural regeneration threatens the persistence of scattered trees, so that they are often referred to as the "living dead" (Fischer et al., 2009; Janzen, 1986, p. 307). Natural regeneration refers to trees establishing from naturally dispersed seed (Dorrough & Moxham, 2005). It is influenced by a range of conditions such as seed supply, soil moisture, exotic vegetation, pathogen attack, competition, fire, elevation, soil fertility, cultivation and grazing (Dorrough & Moxham, 2005; Fischer et al., 2009, Suppl. Fig. S2; Sato

et al., 2016; Vesk & Dorrough, 2006). Variables associated with agricultural intensification such as exotic plant cover, soil fertility and continuous livestock grazing were found to have a negative effect on regeneration (Sato et al., 2016). Consequently, agricultural intensification is an important driver of the regeneration failure and resulting tree decline. Therefore, better management practices need to be found to reconcile production and conservation of scattered trees.

Grazing regimes like rotational grazing and holistic management offer an alternative approach to conventional livestock grazing. In rotational grazing, livestock is moved between multiple paddocks leading to alternating periods of grazing and rest (Heitschmidt & Taylor, 1991, as cited in Briske et al., 2011). Holistic management, also called the Savory grazing method or holistic resource management, involves a form of short-rotational grazing which is characterized by few days of grazing with high stocking rates, followed by extended periods of rest (Savory, 1983; Savory & Parsons, 1980). Instead of a prescribed grazing system, it is a flexible decision-making framework including goal setting, planning procedures for grazing, land, and finances, constant monitoring and adaptation to observed conditions (Savory, 1983; Savory Institute, 2023). For example, the duration of grazing periods is not set, but matched to plant growth rates (Savory, 1983). In this thesis, I use rotational grazing to refer to the narrow ecological definition of alternating periods of grazing and rest, measured as the number of annual grazing days. I use holistic management to refer to Savory's broad definition including the ecological and social dimension.

Rotational grazing and holistic management have been suggested as possible solutions to restore rangelands and promote tree regeneration. Initially, rotational grazing was believed to restore rangelands because the removal of disturbance allows the vegetation to recover (Briske et al., 2011). According to Savory and Parsons (1980), controlling the length of grazing periods rather than stocking rate is what prevents degradation. Furthermore, they postulated that animal impact through hoof action promotes plant succession by breaking up the soil crust and improving water infiltration (Savory & Parsons, 1980). Although research on rotational grazing and holistic management has focused on the effects on soils and ground vegetation (Briske et al., 2008; Savory, 1983; Savory & Parsons, 1980), tree regeneration could be enhanced, too. Grazing influences tree regeneration by trampling, browsing and alteration of soil properties (Windsor, 1999, as cited in Fischer et al., 2009). In rotational grazing regimes trees might recover during rest periods resulting in better regeneration.

However, the ecological benefits of rotational grazing and holistic management are highly contested due to a lack of consistent empirical evidence. The majority of empirical studies did not find rotational grazing strategies to be inherently superior to continuous grazing in terms of their impact on soils, vegetation and livestock (Briske et al., 2008; di Virgilio et al., 2019; Hawkins, 2017; Hawkins et al., 2022; Holechek et al., 2000; Venter et al., 2019). In turn, proponents of holistic management criticised the experiments (Savory, 2013; Savory & Parsons, 1980; Teague & Barnes, 2017) leading to a heated debate about the effectiveness of holistic management which is likely grounded in epistemological

differences between disciplines (Gosnell et al., 2020; Sherren & Kent, 2019). Empirical research on the effect of rotational grazing on trees in particular is sparse and inconsistent. Some studies found a positive effect of shorter grazing periods compared to continuous grazing (Carmona et al., 2013; Fischer et al., 2009; Mayer et al., 2005; Sato et al., 2019), whereas others found no significant difference (Lehmkuhler et al., 1998; McEvoy et al., 2006; Venter et al., 2019). More long-term whole-farm research is needed to draw conclusions about the effects of rotational grazing and holistic management (Teague & Barnes, 2017).

To address this research gap I built on the study by Fischer et al. (2009) who investigated grazing regimes and tree regeneration in the Upper Lachlan Catchment in New South Wales, Australia. In this region, much of the remaining box gum grassy woodland and dry forest occurs as scattered paddock trees (Fischer, Sherren, et al., 2010). Concerning land use, the region is dominated by livestock grazing with some cropping in the west (Fischer, Sherren, et al., 2010). Fischer et al. (2009) found that tree regeneration in the study area is more likely under no grazing or fast-rotational grazing (than under slow-rotational or continuous grazing) and at higher tree densities (i.e. scattered trees are more at risk). This was analysed using tree diameters and seedlings (Fischer et al., 2009). In the long-term the differences in regeneration rates should translate into differences in tree cover between continuous and fast-rotational grazing sites.

In this thesis, I analysed if the predictions by Fischer et al. (2009) hold true by looking at remotely-sensed tree cover changes from 2005 to 2023 in the Upper Lachlan Catchment in relation to grazing regimes and tree densities. Specifically, I tested four hypotheses: (1) Overall, the tree cover has declined in the study area. This assumption is based on the modelled tree decline in the study area by Fischer, Zerger, et al. (2010). (2) The more days grazed per year, the more pronounced the decline in tree cover. Here, annual grazing days are used to measure where a grazing regime can be located on a continuum from fast-rotational to continuous grazing. I expect that less annual grazing days, that means fast-rotational grazing, lead to more tree regeneration and thus cover due to recovery during rest periods. Moreover, Fischer et al. (2009) already found this effect empirically. (3) The relationship between annual grazing days and tree cover decline is more pronounced on sites with scattered trees or grazed woodland. This is because scattered trees are more vulnerable, but in absolute terms, a lot of trees are also lost from sites with higher tree densities (Fischer et al., 2009; Fischer, Zerger, et al., 2010). (4) The decline in tree cover is less pronounced on holistic management farms compared to farms practicing slow-rotational or continuous grazing. This is because holistic managers practice fast-rotational grazing and manage the farm adaptively (Savory, 1983).

Contrary to my expectations, I found that the tree cover may have increased in the study area. There was no significant decline in tree cover with more annual grazing days. The relationship did not differ in the expected way for sites with different tree densities. Tree cover increased slightly on holistic management

and slow-rotation farms, whereas it decreased on continuous farms. Following methods and results, possible reasons for these unexpected results are discussed.

2 METHODS

2.1 STUDY DESIGN

This thesis is a follow-up study to Fischer et al. (2009). For convenience, I provide a brief overview of the study design elements that are most relevant to my research. Detailed information can be found in Fischer et al. (2009). First, the study design was nested. Each farm consisted of several paddocks, some of which contained even smaller study sites (Fig. S5). Second, there were different site types ranging from high to low tree densities: ungrazed woodland, grazed woodland, scattered tree sites, and paddock sites. In addition, there were revegetation sites where farmers actively planted trees. Third, grazing regimes were measured by farm management type (holistic management, slow-rotation, continuous), annual grazing days (in days per year) and stocking rate (in dry sheep equivalents – a 48- to 50-kg wether – per hectare). Holistic management meant graziers had participated in formal training and had started converting to holistic management (Sherren et al., 2012). Annual grazing days showed where a grazing regime was located on a scale from fast-rotational through slow-rotational to continuous grazing. Information on the grazing regime was obtained from farmers by Fischer et al. (2009). I assumed that farmers had not changed their management practices and that these variables were therefore constant over the course of my study. Fourth, Fischer et al. (2009) also mapped the tree cover in 2005 based on SPOT images. Based on the data on grazing regimes and tree cover in 2005 by Fischer et al. (2009) as well as remotely-sensed tree cover in 2023, I calculated the tree cover change from 2005 to 2023 and related it to the grazing regimes.

2.2 TREE COVER MAP 2023 – SUPERVISED CLASSIFICATION

First, I created a 10 m resolution map of the tree cover in 2023 based on satellite images. I retrieved two Sentinel-2 Level-2A (i.e. bottom of the atmosphere reflectance) images with less than 10% cloud cover from Copernicus Open Access Hub (European Space Agency, 2023). Together the images of tiles 55HFB and 55HFC, captured on 11th and 16th February 2023, showed the whole study area in summer. I chose this season because it is less cloudy, and you can better recognise the trees because of the foliage. I pre-processed the images in QGIS (QGIS.org, 2023). Stacking bands 2, 3, 4 and 8 resulted in 10 m resolution 4-band images encompassing blue, green, red, and near-infrared spectral ranges. I combined the two images to one mosaic which was then used to obtain the tree cover (Fig. S5).

I performed a supervised pixel-based classification in ArcGIS Pro (Esri, 2023) to obtain a map of the tree cover in 2023. First, I classified the whole study area into a woody vegetation class and 6 nonwoody classes based on manually created training points (Fig. S6). I used multiple nonwoody classes because the separation of different spectral signatures into different classes increases the accuracy. Afterwards, I aggregated the multiple nonwoody classes to form one nonwoody vegetation class. This reduced the classification to two classes: woody and nonwoody vegetation. Certain parts of the study area were

frequently misclassified because crops or topography led to uncommon spectral signatures. In these regions, I replaced the initial classification with more precise regional classifications each comprising only one or two farms. I tried the whole classification process with two different algorithms. The first one was a Support Vector Machines algorithm (Cortes & Vapnik, 1995) which often produces the most accurate land-cover classifications, the second one was a Maximum Likelihood algorithm which is often used for benchmarking (Khatami et al., 2016). This resulted in two final tree cover maps, one based on the Support Vector Machines algorithm, and one based on the Maximum Likelihood algorithm.

I assessed the accuracy of both maps by comparing the classified values of 500 stratified random points with ground truth values based on the Sentinel-2 image and the World Imagery map (Esri et al., 2023). The Support Vector Machines based classification had an accuracy of 90.0% and a Kappa value of 0.723, whereas the Maximum Likelihood based classification had an accuracy of 93.8% and a Kappa value of 0.816. The confusion matrices showed that the Support Vector Machines overpredicted tree cover more often than the Maximum Likelihood algorithm. Tree cover change values scattered around 0 for both algorithms, that means the tree cover increased in some areas and decreased in others, which seemed reasonable. Anecdotal visual inspection revealed that there was an overprediction of tree cover on a few paddocks. However, most of the study area was classified correctly, and scattered trees were generally captured well. Due to its higher accuracy, I chose the Maximum Likelihood based classification as a map of the tree cover in 2023 (Fig. S7).

2.3 Tree Cover Map 2005

Concerning the tree cover in 2005, I projected the tree cover raster from Fischer et al. (2009) to the same coordinate reference system as the 2023 satellite images. Despite that, the pixels of the 2005 raster did not exactly align with the corresponding pixels of the 2023 raster. Therefore, I used georeferencing in ArcGIS Pro to shift the 2005 tree raster by two pixels which improved the alignment for most of the study area. The result was two comparable tree cover maps for the years 2005 (Fig. S8) and 2023 (Fig. S7).

2.4 Data Cleaning And Tree Cover Change Calculation

Based on the tree cover maps from 2005 and 2023, I created one table each for farms, paddocks and sites containing tree cover and grazing data using Python (Python Software Foundation, 2023). First, I extracted the tree cover on farms, paddocks and sites for both years from the tree cover maps. Second, I calculated the tree cover change from 2005 to 2023 on farms, paddocks and sites by subtraction (see e.g. tree cover change on paddocks in Fig. S9). Third, I cleaned the data to ensure a high quality. I excluded paddocks and sites which were duplicates or contained missing values. I also excluded paddocks and sites where detection of trees was impaired by clouds on the satellite image. Furthermore, I visually inspected outliers, that means paddocks and sites with a tree cover change above the upper quartile plus 1.5 times the interquartile range or below the lower quartile minus 1.5 times the

interquartile range. I excluded outliers which were caused by misclassified pixels rather than actual tree cover changes. Because tree cover on farms was calculated based on the paddocks, all flawed areas were automatically excluded from farm-level data as well. Fourth, I merged the tree cover data on farms, paddocks and sites with the grazing data by Fischer et al. (2009). This resulted in one table each for farms (n = 33), paddocks (n = 1145) and sites (n = 130). The tables contained the tree cover in 2005 and 2023, tree cover change and grazing data (farm management type, annual grazing days and stocking rate).

2.5 STATISTICAL ANALYSIS

To investigate the effect of annual grazing days on tree cover change, I visualised the relationship with scatterplots on paddock- and site-level. I used linear mixed effect models (LME) (Pinheiro & Bates, 2000) to test for statistically significant effects of annual grazing days on tree cover change on paddock- and site-level. I included the farm as a random effect. Furthermore, I used a linear mixed effect model to test for statistically significant effects of the interaction of annual grazing days and site type on tree cover change on site-level, while accounting for the random effect of the farm.

To investigate differences in tree cover change between farm management types, I visualised them with boxplots on farm-, paddock- and site-level. I used analysis of variance (ANOVA) to test for statistically significant differences in mean tree cover change between farm management types on farm-level. I used linear mixed effect models to test for statistically significant differences in mean tree cover change between farm management types on paddock- and site-level, while accounting for the random effect of the farm.

On paddock- and site-level, I performed all analyses using only grazed paddocks and sites (excluding ungrazed and revegetated ones). Farms on the other hand did contain ungrazed and revegetated areas alongside grazed areas. All statistical analyses were performed in R (R Core Team, 2024) using the packages “nlme” (Pinheiro et al., 2023) for linear mixed effect models and “ggplot2” (Wickham, 2016) and “wesanderson” (Ram & Wickham, 2023) for graphics.

3 RESULTS

3.1 TREE COVER CHANGE

Tree cover increased by 4.0 percentage points on the average paddock in the study area. However, on grazed paddocks (excluding revegetated and ungrazed paddocks), the mean increase was only 0.8 percentage points. Revegetation sites had the highest increase in tree cover, followed by grazed woodlands (Fig. 1). Tree cover stayed approximately the same in ungrazed woodlands, scattered tree sites and paddock sites (Fig. 1).

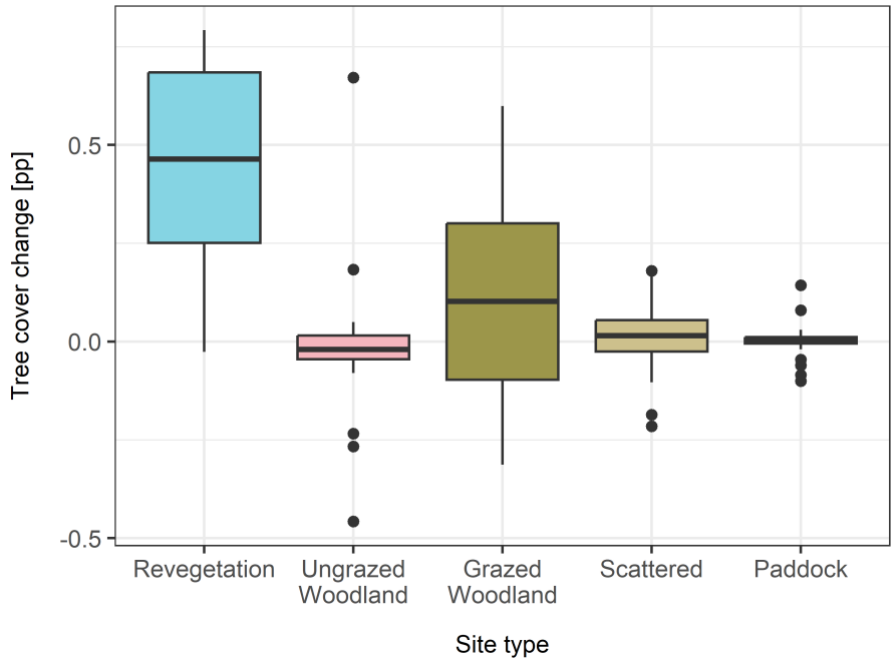


Figure 1. Boxplots of tree cover change [percentage points] for different site types.

3.2 EFFECT OF ANNUAL GRAZING DAYS ON TREE COVER CHANGE

Annual grazing days did not have a significant effect on tree cover change on paddocks (LME, $p = 0.166$; Fig. 2). Because model assumptions were violated, I only graphically investigated all patterns on site level. Tree cover change did not seem to be related to annual grazing days on sites (Fig. S1).

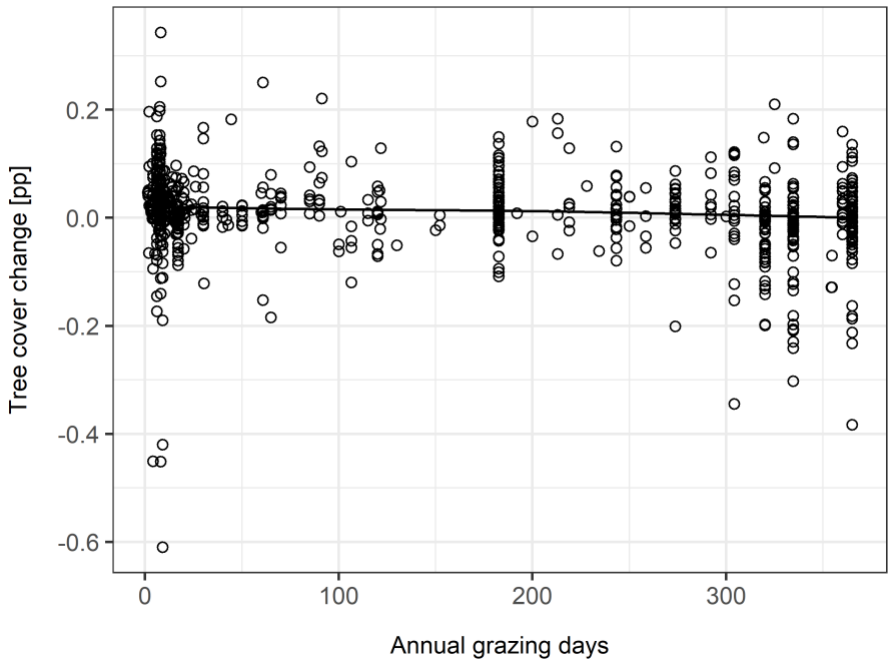


Figure 2. Scatterplot (with loess-curve) of tree cover change [percentage points] in relation to annual grazing days on paddocks.

3.3 Effect of Tree Density on the Relation of Annual Grazing Days and Tree Cover Change

Next, I looked at tree cover change in relation to annual grazing days on sites with different tree densities. On scattered and paddock sites, tree cover change did not seem to be related to annual grazing days (Fig. 3). On grazed woodland sites, the tree cover seemed to increase on sites with a high number of annual grazing days, but there was very high variance (Fig. 3).

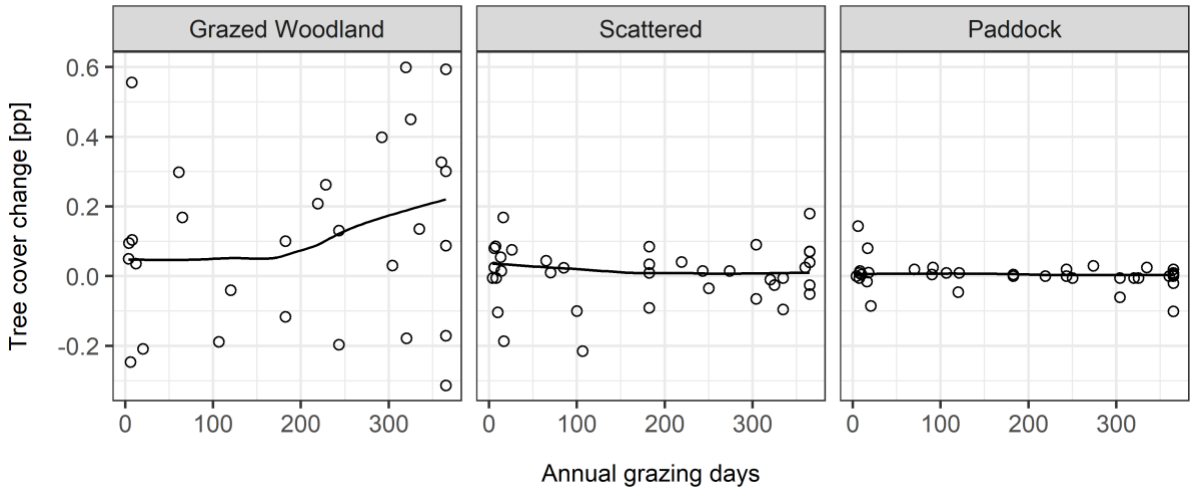


Figure 3. Scatterplots (with loess-curves) of tree cover change [percentage points] in relation to annual grazing days on grazed woodland, scattered and paddock sites.

3.4 Effect of Farm Management Type on Tree Cover Change

Finally, I analysed the effect of farm management type on tree cover change. Tree cover increased slightly on holistic management and slow-rotation farms, whereas it decreased on continuous farms (Fig. S2), but the differences were not statistically significant (ANOVA, $p = 0.253$). Tree cover increased slightly on paddocks belonging to holistic management and slow-rotation farms, whereas it decreased on paddocks belonging to continuous farms (Fig. 4). The difference in mean tree cover change between holistic management and continuous paddocks was statistically significant (LME, $p = 0.018$). There was no statistically significant difference in tree cover change between holistic management and slow-rotation paddocks (LME, $p = 0.471$). On sites there was no visible difference in tree cover change between different management types (Fig. S3).

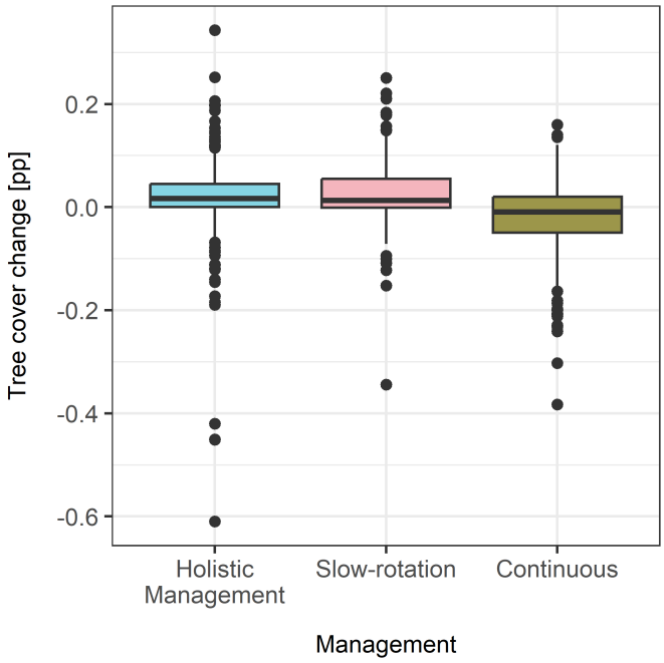


Figure 4. Boxplots of tree cover change [percentage points] on paddocks belonging to farms with different management types.

4 DISCUSSION

4.1 TREE COVER CHANGE

My tree cover change analysis revealed that tree cover may have increased in the study area overall. However, most of the increase is likely caused by revegetation and not by natural regeneration. Tree cover may have slightly increased on grazed paddocks as well. Therefore, I did not find evidence for the hypothesised tree cover decline. This is surprising given Fischer, Zerger, et al. (2010) predicted a tree cover decline for the area that should be visible within the studied time period. There are two possible explanations for these unexpected results. Either the tree cover actually increased or methodological limitations distorted the results.

Tree cover could have increased if there was more rainfall in recent years which supports regeneration. Vesk and Dorrough (2006) found that rainfall, together with seed supply, had a high impact on natural regeneration leading to a high variability in regeneration. Both annual and seasonal rainfall in New South Wales in recent years followed a similar pattern to previous years (Australian Government Bureau of Meteorology, 2024a). Rainfall in the study area, that means around Cowra more specifically, could still have increased. The Australian Government Bureau of Meteorology (2024b) only provides monthly summary statistics and no easily accessible time series data on annual and seasonal rainfall in Cowra. Therefore, locally higher rainfall might have contributed to the tree cover increase, but this needs further investigation.

Another possible cause for the tree cover increase is the Environmental Stewardship Program (Australian Government Department of Climate Change, Energy, the Environment and Water, n.d.). The program included a box gum grassy woodland project which provided land managers with funding to improve their patches of box gum grassy woodland, at least 10 hectares in size, for 10 to 15 years starting 2010 (Australian Government Department of Climate Change, Energy, the Environment and Water, n.d.). Required management actions included retaining all standing trees, managing regeneration and other actions that could promote tree regeneration such as strategic grazing (Australian Government Department of Climate Change, Energy, the Environment and Water, n.d.). Many of the funded patches are located in my study area and could overlap with the paddocks I analysed (Environmental Resources Information Network, 2014). Because the measures were established after the study by Fischer et al. (2009), any paddock, including the ones I classified as grazed, might be influenced by the new measures. This could have led to enhanced regeneration in the study area explaining the unexpected tree cover increase even on grazed paddocks.

The other option is that tree cover did decline, but I could not detect it due to methodological limitations. The increasing tree cover on revegetation sites shows that the classification must be roughly correct and can be used to compare tree cover change between sites with different conditions. However, the remote sensing techniques used are not well suited to quantify absolute changes over time. The satellite and

classification algorithm I used for mapping tree cover in 2023 differed from those used in 2005. Moreover, neither of the classifications were completely accurate. Even a small overprediction of tree cover in 2023 could lead to a perceived tree cover increase, especially if the classification in 2005 was more reluctant to classify a pixel as a tree. Although I cannot reject the null hypothesis of no tree cover decline for now, I would be cautious about relying on these results.

Instead, I recommend continuing to monitor tree cover change because we should not risk losing scattered trees. Due to their role as keystone structures, a loss of scattered trees would entail cascading negative effects on the ecosystem and the people depending on it (Fischer, Zerger, et al., 2010; Manning et al., 2006). Future monitoring efforts should ideally use higher resolution satellite images with more accurate and consistent classification algorithms or field surveys to overcome the limitations that I faced.

4.2 Effect of Annual Grazing Days on Tree Cover Change

Contrary to my second hypothesis, annual grazing days did not show an effect on tree cover change. This is in line with Lehmkuhler et al. (1998) who found that tree diameter and height growth in mature walnut stands were not significantly different between continuous and rotational cattle grazing. Similarly, height and canopy diameter of similar-aged oak saplings were not significantly different between continuous and rotational pony grazing in Spain (McEvoy et al., 2006). However, both studies had only two replicates of each grazing system and lasted no longer than a year (Lehmkuhler et al., 1998; McEvoy et al., 2006). Since some grazing effects can take several years to be expressed (Öllerer et al., 2019), it is not surprising that they did not find significant differences. Another study that aligns with my findings is by Venter et al. (2019) who analysed remotely-sensed vegetation across farm fence-line contrasts and also found no significant differences in tree cover between rotational and continuous grazing. Consequently, it is common that no evidence for an effect of rotational grazing is found, and my study is no exception.

On the other hand, my results are surprising when seen in relation to the theory by Savory and Parsons (1980) combined with non-equilibrium theory. Savory and Parsons (1980) stated that mimicking the behaviour of wild herds with short periods of high animal impact followed by extended rest positively impacts vegetation. Non-equilibrium theory indicates that this is especially true in areas of high rainfall variability. Where rainfall variability is high (coefficient of variation > 33%), degradation only occurs in the presence of water or key resources (von Wehrden et al., 2012). This is because wild herbivore populations collapse during droughts allowing the vegetation to recover (von Wehrden et al., 2012). Artificially stabilizing animal numbers in areas with high rainfall variability by supplying additional water and food in the dry season can lead to severe degradation (Illius & O'Connor, 2000). Rotational grazing on the other hand includes rest periods that can reproduce the periodicity in herbivore populations (von Wehrden et al., 2012). My study area has a medium high rainfall variability with a coefficient of variation around 30% (von Wehrden et al., 2012, Suppl.). Since rotational grazing was postulated to be superior to continuous grazing in general and even more so in areas of high rainfall

variability, I would expect less annual grazing days to benefit the trees in my study area. Therefore, it is surprising that I did not find an effect of annual grazing days on tree cover change.

My results not only contradict the underlying theory, but also the empirical studies in which an effect of the length of the grazing period on tree regeneration was observed. Carmona et al. (2013) found that holm oak juveniles in Mediterranean dehesas were less browsed, had bigger canopy cover and higher sapling density under seasonal grazing compared to year-long continuous grazing. On subalpine wood pastures, Mayer et al. (2005) observed an increase in the percentage of browsing damage with the length of the grazing period. Due to different study areas, measurement methods, tree species and rotational grazing systems, differences to my results are plausible. More surprising is the contradiction of my results and those of Fischer et al. (2009) because they investigated the same study area with a similar study design. Fischer et al. (2009) found that the presence of seedlings was more likely under no grazing or fast-rotational grazing (than under slow-rotational or continuous grazing). Looking at the same study area 17 years later should reveal a relation of tree cover to annual grazing days which represent a continuum from fast-rotational to continuous grazing. However, I did not find such a relationship.

Interestingly, Sato et al. (2019) also did not find a significant effect on overstorey cover, but did find one on regeneration in the same study area. More specifically, past (i.e. pre-2011) grazing regime (continuous, short-conversion rotational, long-conversion rotational) was not significantly associated with tree regeneration and overstorey cover (Sato et al., 2019). This is in line with my results. With increasing duration of present (i.e. post-2011) grazing events, sites were significantly less likely to support native tree regeneration (Sato et al., 2019). This is in line with the results of Fischer et al. (2009). These findings suggest that looking at tree cover as opposed to seedlings is the main cause of differing results.

One possible reason is that the differences in seedling presence between rotational and continuous grazing regimes might be levelled out in the sapling or young tree stage resulting in a lack of differences in tree cover. However, this is unlikely to be true. Vesk and Dorrough (2006) considered eucalypt saplings above 1 m height to have escaped because they would be little affected by grazing at this stage. Carmona et al. (2013) suggested that saplings could still be damaged by grazing. Even then, to level out the effect more saplings needed to die on rotational grazing paddocks than on continuous ones and there is no reason to assume that. On the contrary, Carmona et al. (2013) found more old saplings on seasonally grazed sites than on continuous year-long grazed sites. Therefore, it is unlikely that sapling deaths cause the discrepancy between results seen in seedlings compared to tree cover.

A different reason for the unexpected lack of an effect of annual grazing days on tree cover could be that any existing effect is masked by methodological limitations. First, classification errors in 2005 and 2023 could conceal the effect. Secondly, tree crowns of naturally regenerated trees might not be visible on the 10 m resolution satellite imagery, yet. This applies in particular to trees that have only recently germinated and are still in the seedling stage. Furthermore, old trees might not have died yet, so that the

change seen on satellite images is minimal. Thirdly, the box gum grassy woodland project might have led to inconsistent grazing regimes interfering with my assumption of consistent grazing regimes. Paddocks I thought had a high number of annual grazing days might have been converted to ungrazed or rotationally grazed paddocks with low numbers of annual grazing days as part of the strategies of the project. This would distort the results. However, Sato et al. (2019) did consider the altered grazing practices caused by the project and did not find an effect on tree cover either.

Fourth and most importantly, there could be too many confounding factors that mask any real pattern. Natural regeneration is influenced by a range of conditions such as seed supply, soil moisture, exotic vegetation, pathogen attack, competition, fire, elevation, soil fertility, cultivation and grazing (Dorrough & Moxham, 2005; Fischer et al., 2009, Suppl. Fig. S2; Sato et al., 2016; Vesk & Dorrough, 2006). Annual grazing days is only one of them. Location-based factors such as the rainfall gradient across the study area were probably not systematically related to annual grazing days. Although a bit more paddocks with few annual grazing days were in the west, paddocks with the same number of annual grazing days were distributed throughout the study area. Even if there is no systematic impact on the results, these factors still introduce much variance. Some of them, namely rainfall and seed variability and stocking rate, were proposed to have overwhelming effects (Briske et al., 2008; Vesk & Dorrough, 2006). I graphically checked if low (up to 5 dry sheep equivalent per hectare) compared to high stocking rates (> 5 dry sheep equivalent per hectare) influenced the relation between annual grazing days and tree cover change, which did not seem to be the case. Furthermore, any influence of the different farms was controlled for by including the farm as a random effect in the model. Apart from that, no potentially confounding factors were controlled for in my study which means they could interfere with the results. Sato et al. (2019) included more factors in their model such as fertilization, stocking rate, natural soil fertility, native woody cover and mean annual rainfall. They observed a lack of improvement in woodland condition despite rotational grazing because a multitude of factors influenced the effectiveness of interventions (Sato et al., 2019). This confirms that trees are influenced by many factors which potentially masked an effect of annual grazing days on tree cover in my results.

To summarize, one can see an effect of rotational grazing on seedlings, but not on tree cover. Both Fischer et al. (2009) and Sato et al. (2019) observed a significant relationship between grazing regimes and eucalypt regeneration in New South Wales. Both me and Sato et al. (2019) did, however, not find an effect on tree cover. It is unclear whether this accurately reflects reality due to multiple limitations of my study. Therefore, I can neither accept nor reject the null hypothesis that tree cover decline is unrelated to annual grazing days. Based on the reflections above, I find it more convincing that there is an effect that is just not visible in the tree cover yet due to methodological limitations. However, more research is needed to clarify this.

4.3 Effect of Tree Density on the Relation of Annual Grazing Days and Tree Cover Change

Concerning the effect of tree density on the relation between annual grazing days and tree cover change, there was no graphical relation, contrary to my hypothesis. There was only one visible pattern. In grazed woodlands, tree cover seemed to increase on sites with a high number of annual grazing days. There are two possible explanations for this. Possibility one is that longer grazing periods actually lead to more tree growth in grazed woodlands, contrary to Savory and Parsons' (1980) theory. Grazing does not always have purely negative effects on vegetation. In some cases, it can also have positive ones. For example, the removal of competitive vegetation can benefit tree regeneration (Öllerer et al., 2019). However, considering the results of Fischer et al. (2009) and Sato et al. (2019), a positive effect of more grazing days in grazed woodlands in this study area is unlikely.

Possibility two is that a few "grazed woodland" paddocks have been fenced and turned into ungrazed paddocks or planted with trees leading to a higher tree cover increase. That this happens specifically on paddocks with a higher number of annual grazing days can be explained by the fact that conventional graziers tend to have a land sparing approach (Sherren et al., 2012). That means they use one part of the land for agriculture and set aside another part for conservation (Sherren et al., 2012). By contrast, graziers practicing holistic management with low numbers of annual grazing days tend to integrate production and conservation in the same area (land sharing) (Sherren et al., 2012). The box gum grassy woodland project encourages these sudden changes to paddock management by paying graziers for actively restoring patches of their land. This seems to be the more plausible explanation for a tree cover increase on grazed woodland patches with high annual grazing days. However, the relation should be viewed with caution because it could not be tested for significance.

4.4 Effect of Farm Management Type on Tree Cover Change

In line with my fourth hypothesis, tree cover increased slightly on holistic management and slow-rotation farms and paddocks but decreased on continuous ones. Farm-level data included revegetated areas, whereas paddock-level data excluded revegetated areas. In spite of that, the same pattern was observed on both levels. This indicates that revegetation was not the reason for differences between farm management types. The differences in tree cover change between farm management types were very small (Fig. 4). The difference between continuous and holistic management was not significant on farm-level but was significant on paddock-level. This could be due to the bigger sample size of paddocks ($n = 1145$) compared to farms ($n = 33$) which increases the probability that significant differences from chance are found. On site-level there were no visible differences between farm management types at all (Fig. S3). These inconsistencies suggest that the effect is very small and there is high variance. Consequently, there are probably many other important factors that I did not include that led to unexplained variance. The other limitations mentioned in the section about annual grazing days also

apply to farm management types and could further mask the results. Still, there is a small significant effect of farm management type on paddock-level.

It is surprising that there is a small effect of farm management type even though there is no effect of annual grazing days because both are strongly linked. Paddocks of holistic management farms usually have few annual grazing days and those of continuous farms have high numbers of annual grazing days (Fig. S4). There are only few outlier paddocks that do not conform to this pattern. These outliers might affect the results strongly enough that the effect of farm management type is significant on paddock-level even though annual grazing days are not.

The difference in results between annual grazing days and farm management type might be caused by other ecological factors included in holistic management. Next to fast-rotational grazing, holistic management in Australia often involves reduced chemical fertiliser use and native pastures (Sherren et al., 2012). Conventional graziers on the other hand often practice continuous grazing, use exotic or annual pastures and regularly apply chemical fertiliser (Sherren et al., 2012). Observed differences between holistic management and conventional farming might not be primarily caused by annual grazing days. They could just as well be caused by one of the other ecological factors that are linked to farm management or by their sum. Since I did not consider these factors in my analysis, any of these possibilities could apply. Another ecological factor being the main driver for tree cover changes could therefore explain why annual grazing days were not significant, but farm management type was.

The difference in results between annual grazing days and farm management type might also be caused by the social dimension of holistic management. It is interesting that the broader concept of holistic management showed an effect, whereas the narrow ecological measure of annual grazing days did not. This resembles the contrast between experimental studies and social science studies which more often found positive effects of holistic management (Sherren & Kent, 2019). An important part of holistic management is its integration of elements of systems thinking such as interdisciplinarity, holism, feedback loops and adaptivity (Mann & Sherren, 2018). Sherren et al. (2012) provided first evidence for the theory that holistic management graziers practice more adaptive management and think more systematically about their grazing regime. For example, conventional graziers see a drought as a temporary exception that they need to endure, whereas holistic management graziers adapt by monitoring ground cover and changing stocking rates (Sherren et al., 2012). Later, Mann et al. (2019) corroborated the findings. Higher adherence to holistic management correlated with stronger systems thinking (Mann et al., 2019). Benefits of holistic management might be caused by increased systems thinking, management attention and adaptive decision-making instead of fast-rotational grazing alone (Briske et al., 2011; Sherren et al., 2012). Unfortunately, this remains speculative until further research is conducted.

More research on the effects of rotational grazing and holistic management is needed because they are still poorly understood and highly contested. Since the debate is largely rooted in disciplinary

differences, any future research on these grazing regimes should be inter- or transdisciplinary (Sherren & Kent, 2019; Teague & Barnes, 2017). However, more research alone is unlikely to resolve the controversy because there is no consensus on what constitutes good research among the parties (Gosnell et al., 2020). A different option is to move beyond the debate of which grazing regime is better by focussing on common ground. An aspect both sides of the debate seem to agree on is that holistic management fosters systems thinking and adaptive decision-making which could lead to better management effectiveness and indirect ecological benefits (Briske et al., 2011; Gosnell et al., 2020; Sherren et al., 2012). Sherren et al. (2012) take this as a reason to advocate for holistic management, whereas Briske et al. (2011) advocate for adaptive management regardless of the grazing regime. A promising avenue for future research would be to assess the proposed effects of adaptive management, including its effect on tree regeneration. Furthermore, it could be analysed if the benefits of adaptive management can be achieved with any grazing regime or with holistic management only.

For practitioners, my results show that the biggest increase in tree cover was achieved by planting trees. Although this is more effective in the short term, previous research suggested that it is not enough. Tree planting is mostly done on larger patches, but it is important that scattered trees in particular are retained due to their unique ecological functions such as increased landscape connectivity (Manning et al., 2006). Therefore, natural regeneration of scattered trees needs to be supported. Reduced fertilization and fast-rotational grazing might be beneficial for natural regeneration and woodland condition (Fischer et al., 2009; Sato et al., 2019), but there is still no conclusive evidence on their translation into more tree cover. Furthermore, holistic management might be good because it is linked to systems thinking and adaptive management. These practices might be the best shot at promoting natural regeneration and preserving scattered trees until researchers find more promising approaches.

5 CONCLUSION

Following up on Fischer et al. (2009), I analysed remotely-sensed tree cover changes from 2005 to 2023 in the Upper Lachlan Catchment and their relation to tree density, rotational grazing and holistic management. First, I found that the tree cover may have increased in the study area contrary to my expectations. This could have been supported by increased rainfall in the study area and by the box gum grassy woodland project which funded restoration efforts. However, these results should be treated with caution because the remote sensing techniques used are not ideal for quantifying changes over time and might not produce accurate results. Second, there was no significant decline in tree cover with more annual grazing days. This is surprising in light of Savory and Parsons' (1980) theory and Fischer et al. (2009) and Sato et al.'s (2019) results showing an effect of grazing regimes on eucalypt regeneration. Consequently, there seems to be an effect on seedlings, but not on tree cover. It might be the case that the effect is not observable in the tree cover yet due to multiple methodological limitations. Third, the relationship between annual grazing days and tree cover did not differ in the expected way for sites with different tree densities. In grazed woodlands, tree cover seemed to increase on sites with a high number of annual grazing days which could be due to grazing cessation or tree planting. Fourth, tree cover increased slightly on holistic management and slow-rotation farms, whereas it decreased on continuous farms. This effect was only significant on paddock-level which suggests that the effect is very small. It is surprising that there is a small effect of farm management type although there is no effect of annual grazing days. The difference might be caused by confounding ecological factors or systems thinking and adaptive management of holistic managers.

Avenues for future research may be further monitoring of scattered tree cover change and the analysis of effects of adaptive management, including its effect on tree regeneration. Although the evidence is inconclusive, for now, the best chance of conserving scattered trees might lie in the adoption of holistic management including fast-rotational grazing, reduced fertilization and adaptive decision-making.

6 REFERENCES

Australian Government Bureau of Meteorology. (2024a). *Australian climate variability & change—Time series graphs*. Last accessed 24 June 2024 at http://www.bom.gov.au/cgi-bin/climate/change/timeseries.cgi?graph=rain&area=nsw&season=0112&ave_yr=0

Australian Government Bureau of Meteorology. (2024b, June 20). *Climate statistics for Australian locations*. Last accessed 24 June 2024 at http://www.bom.gov.au/climate/averages/tables/cw_065111.shtml

Australian Government Department of Climate Change, Energy, the Environment and Water. (n.d.). *Environmental Stewardship—Box Gum Grassy Woodland Project—Information Booklet*. Last accessed 24 June 2024 at https://www.dcceew.gov.au/sites/default/files/documents/environmental-stewardship-info-booklet-box-gum-grassy-woodland-project.pdf

Briske, D. D., Derner, J. D., Brown, J. R., Fuhlendorf, S. D., Teague, W. R., Havstad, K. M., Gillen, R. L., Ash, A. J., & Willms, W. D. (2008). Rotational Grazing on Rangelands: Reconciliation of Perception and Experimental Evidence. *Rangeland Ecology & Management*, *61*(1), 3–17. https://doi.org/10.2111/06-159R.1

Briske, D. D., Sayre, N. F., Huntsinger, L., Fernandez-Gimenez, M., Budd, B., & Derner, J. D. (2011). Origin, Persistence, and Resolution of the Rotational Grazing Debate: Integrating Human Dimensions Into Rangeland Research. *Rangeland Ecology & Management*, *64*(4), 325–334. https://doi.org/10.2111/REM-D-10-00084.1

Carmona, C. P., Azcárate, F. M., Oteros-Rozas, E., González, J. A., & Peco, B. (2013). Assessing the effects of seasonal grazing on holm oak regeneration: Implications for the conservation of Mediterranean dehesas. *Biological Conservation*, *159*, 240–247. https://doi.org/10.1016/j.biocon.2012.11.015

Cortes, C., & Vapnik, V. (1995). Support-vector networks. *Machine Learning*, *20*(3), 273–297. https://doi.org/10.1007/BF00994018

di Virgilio, A., Lambertucci, S. A., & Morales, J. M. (2019). Sustainable grazing management in rangelands: Over a century searching for a silver bullet. *Agriculture, Ecosystems & Environment*, *283*, 106561. https://doi.org/10.1016/j.agee.2019.05.020

Dorrough, J., & Moxham, C. (2005). Eucalypt establishment in agricultural landscapes and implications for landscape-scale restoration. *Biological Conservation*, *123*(1), 55–66. https://doi.org/10.1016/j.biocon.2004.10.008

England, J. R., O'Grady, A. P., Fleming, A., Marais, Z., & Mendham, D. (2020). Trees on farms to support natural capital: An evidence-based review for grazed dairy systems. *Science of The Total Environment*, *704*, 135345. https://doi.org/10.1016/j.scitotenv.2019.135345

Environmental Resources Information Network. (2014). *Environmental Stewardship Programme Projects* [Map]. Australian Government Department of the Environment. Last accessed 24 June 2024 at https://www.dcceew.gov.au/sites/default/files/documents/esp-project-map.pdf

Esri. (2023). *ArcGIS Pro* (Version 3.1.3) [Computer software]. https://www.esri.com/de-de/arcgis/products/arcgis-pro/overview

Esri, HERE, Garmin, OpenStreetMap contributors, & GIS User Community. (2011). *World Light Gray Base* [Basemap]. https://www.arcgis.com/home/item.html?id=ed712cb1db3e4bae9e85329040fb9a49

Esri, Maxar, Earthstar Geographics, & GIS User Community. (2023). *World Imagery* [Basemap]. https://www.arcgis.com/home/item.html?id=10df2279f9684e4a9f6a7f08febac2a9

Esri, TomTom, Garmin, FAO, NOAA, USGS, OpenStreetMap contributors, & GIS User Community. (2017). *Hybrid Reference Layer* [Basemap]. https://www.arcgis.com/home/item.html?id=30d6b8271e1849cd9c3042060001f425

European Space Agency. (2023). *Copernicus Open Access Hub*. Last accessed 24 June 2024 at https://scihub.copernicus.eu/

Fischer, J., Sherren, K., Stott, J., Zerger, A., Warren, G., & Stein, J. (2010). Toward landscape-wide conservation outcomes in Australia's temperate grazing region. *Frontiers in Ecology and the Environment*, *8*(2), 69–74. https://doi.org/10.1890/080170

Fischer, J., Stott, J., & Law, B. S. (2010). The disproportionate value of scattered trees. *Biological Conservation*, *143*(6), 1564–1567. https://doi.org/10.1016/j.biocon.2010.03.030

Fischer, J., Stott, J., Zerger, A., Warren, G., Sherren, K., & Forrester, R. I. (2009). Reversing a tree regeneration crisis in an endangered ecoregion. *Proceedings of the National Academy of Sciences*, *106*(25), 10386–10391. https://doi.org/10.1073/pnas.0900110106

Fischer, J., Zerger, A., Gibbons, P., Stott, J., & Law, B. S. (2010). Tree decline and the future of Australian farmland biodiversity. *Proceedings of the National Academy of Sciences*, *107*(45), 19597–19602. https://doi.org/10.1073/pnas.1008476107

Gibbons, P., Lindenmayer, D. B., Fischer, J., Manning, A. D., Weinberg, A., Seddon, J., Ryan, P., & Barrett, G. (2008). The Future of Scattered Trees in Agricultural Landscapes. *Conservation Biology*, *22*(5), 1309–1319. https://doi.org/10.1111/j.1523-1739.2008.00997.x

Gosnell, H., Grimm, K., & Goldstein, B. E. (2020). A half century of Holistic Management: What does the evidence reveal? *Agriculture and Human Values*, *37*(3), 849–867. https://doi.org/10.1007/s10460-020-10016-w

Hawkins, H.-J. (2017). A global assessment of Holistic Planned Grazing™ compared with season-long, continuous grazing: Meta-analysis findings. *African Journal of Range & Forage Science*, *34*(2), 65–75. https://doi.org/10.2989/10220119.2017.1358213

Hawkins, H.-J., Venter, Z.-S., & Cramer, M. D. (2022). A holistic view of Holistic Management: What do farm-scale, carbon, and social studies tell us? *Agriculture, Ecosystems & Environment*, *323*, 107702. https://doi.org/10.1016/j.agee.2021.107702

Holechek, J., Gomes, H., Molinar, F., Galt, D., & Valdez, R. (2000). Short-Duration Grazing: The Facts in 1999. *Rangelands*, *22*, 18–22. https://doi.org/10.2458/azu_rangelands_v22i1_holechek

Illius, A. W., & O'Connor, T. G. (2000). Resource heterogeneity and ungulate population dynamics. *Oikos*, *89*(2), 283–294. https://doi.org/10.1034/j.1600-0706.2000.890209.x

Janzen, D. H. (1986). The Future of Tropical Ecology. *Annual Review of Ecology and Systematics*, *17*, 305–324.

Khatami, R., Mountrakis, G., & Stehman, S. V. (2016). A meta-analysis of remote sensing research on supervised pixel-based land-cover image classification processes: General guidelines for practitioners and future research. *Remote Sensing of Environment*, *177*, 89–100. https://doi.org/10.1016/j.rse.2016.02.028

Lehmkuhler, J. W., Kerley, M. S., Garrett, H. E., Cutter, B. E., & McGraw, R. L. (1998). Comparison of continuous and rotational silvopastoral systems for established walnut plantations in southwest Missouri, USA. *Agroforestry Systems*, *44*(2), 267–279. https://doi.org/10.1023/A:1006206929983

Mann, C., Parkins, J. R., Isaac, M. E., & Sherren, K. (2019). Do practitioners of holistic management exhibit systems thinking? *Ecology and Society*, *24*(3). https://www.jstor.org/stable/26796984

Mann, C., & Sherren, K. (2018). Holistic Management and Adaptive Grazing: A Trainers' View. *Sustainability*, *10*(6), Article 6. https://doi.org/10.3390/su10061848

Manning, A. D., Fischer, J., & Lindenmayer, D. B. (2006). Scattered trees are keystone structures – Implications for conservation. *Biological Conservation*, *132*(3), 311–321. https://doi.org/10.1016/j.biocon.2006.04.023

Manning, A. D., Gibbons, P., & Lindenmayer, D. B. (2009). Scattered trees: A complementary strategy for facilitating adaptive responses to climate change in modified landscapes? *Journal of Applied Ecology*, *46*(4), 915–919. https://doi.org/10.1111/j.1365-2664.2009.01657.x

Mayer, A. C., Estermann, B. L., Stöckli, V., & Kreuzer, M. (2005). Experimental determination of the effects of cattle stocking density and grazing period on forest regeneration on a subalpine wood pasture. *Animal Research*, *54*(3), 153–171. https://doi.org/10.1051/animres:2005018

McEvoy, P. M., McAdam, J. H., Mosquera-Losada, M. R., & Rigueiro-Rodríguez, A. (2006). Tree Regeneration and Sapling Damage of Pedunculate Oak Quercus robur in a Grazed Forest in Galicia, NW Spain: A Comparison of Continuous and Rotational Grazing Systems. *Agroforestry Systems*, *66*(2), 85–92. https://doi.org/10.1007/s10457-005-2916-0

Öllerer, K., Varga, A., Kirby, K., Demeter, L., Biró, M., Bölöni, J., & Molnár, Z. (2019). Beyond the obvious impact of domestic livestock grazing on temperate forest vegetation – A global review. *Biological Conservation*, *237*, 209–219. https://doi.org/10.1016/j.biocon.2019.07.007

Pinheiro, J. C., & Bates, D. M. (2000). *Mixed-Effects Models in S and S-PLUS*. Springer-Verlag. https://doi.org/10.1007/b98882

Pinheiro, J. C., Bates, D. M., & R Core Team. (2023). *nlme: Linear and Nonlinear Mixed Effects Models* (Version 3.1-164) [R package]. https://CRAN.R-project.org/package=nlme

Prevedello, J. A., Almeida-Gomes, M., & Lindenmayer, D. B. (2018). The importance of scattered trees for biodiversity conservation: A global meta-analysis. *Journal of Applied Ecology*, *55*(1), 205–214. https://doi.org/10.1111/1365-2664.12943

Python Software Foundation. (2023). *Python* (Version 3.8.18) [Computer software]. https://www.python.org/

QGIS.org. (2023). *QGIS Geographic Information System.* [Computer software]. QGIS Association. https://www.qgis.org/en/site/

R Core Team. (2024). *R: A Language and Environment for Statistical Computing* (Version 4.3.3) [Computer software]. R Foundation for Statistical Computing, Vienna, Austria. https://www.r-project.org/

Ram, K., & Wickham, H. (2023). *wesanderson: A Wes Anderson Palette Generator* (Version 0.3.7) [R package]. https://CRAN.R-project.org/package=wesanderson

Sato, C. F., Florance, D., & Lindenmayer, D. B. (2019). Drivers of temperate woodland condition through time in an agricultural landscape. *Land Degradation & Development*, *30*(11), 1357–1367. https://doi.org/10.1002/ldr.3325

Sato, C. F., Wood, J. T., Stein, J. A., Crane, M., Okada, S., Michael, D. R., Kay, G. M., Florance, D., Seddon, J., Gibbons, P., & Lindenmayer, D. B. (2016). Natural tree regeneration in agricultural landscapes: The implications of intensification. *Agriculture, Ecosystems & Environment*, *230*, 98–104. https://doi.org/10.1016/j.agee.2016.05.036

Savory, A. (1983). The Savory Grazing Method or Holistic Resource Management. *Rangelands*, *5*, 155–159.

Savory, A. (2013). *Response to Request for Information on the "Science" and "Methodology" Underpinning Holistic Management and Holistic Planned Grazing*. https://savory.global/wp-content/uploads/2018/08/The_Science_and_Methodolgy_of_Holistic_Planned_Grazing.pdf

Savory, A., & Parsons, S. D. (1980). The Savory grazing method. *Rangelands*, *2*, 234–237.

Savory Institute. (2023). *Holistic Management*. Last accessed 24 June 2024 at https://savory.global/holistic-management/

Sherren, K., Fischer, J., & Fazey, I. (2012). Managing the grazing landscape: Insights for agricultural adaptation from a mid-drought photo-elicitation study in the Australian sheep-wheat belt. *Agricultural Systems*, *106*(1), 72–83. https://doi.org/10.1016/j.agsy.2011.11.001

Sherren, K., & Kent, C. (2019). Who's afraid of Allan Savory? Scientometric polarization on Holistic Management as competing understandings. *Renewable Agriculture and Food Systems*, *34*(1), 77–92. https://doi.org/10.1017/S1742170517000308

Teague, W. R., & Barnes, M. (2017). Grazing management that regenerates ecosystem function and grazingland livelihoods. *African Journal of Range & Forage Science*, *34*, 1–10. https://doi.org/10.2989/10220119.2017.1334706

Venter, Z. S., Cramer, M. D., & Hawkins, H.-J. (2019). Rotational grazing management has little effect on remotely-sensed vegetation characteristics across farm fence-line contrasts. *Agriculture, Ecosystems & Environment*, *282*, 40–48. https://doi.org/10.1016/j.agee.2019.05.019

Vesk, P., & Dorrough, J. (2006). Getting trees on farms the easy way? Lessons from a model of eucalypt regeneration on pastures. *Australian Journal of Botany*, *54*. https://doi.org/10.1071/BT05188

von Wehrden, H., Hanspach, J., Kaczensky, P., Fischer, J., & Wesche, K. (2012). Global assessment of the non-equilibrium concept in rangelands. *Ecological Applications*, *22*(2), 393–399. https://doi.org/10.1890/11-0802.1

Wickham, H. (2016). *ggplot2: Elegant Graphics for Data Analysis* (Second edition). Springer-Verlag. https://doi.org/10.1007/978-3-319-24277-4

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FIGURE S6. CLASSIFIED MAP OF LAND COVER IN 2023 USING A WOODY AND 6 NON-WOODY CLASSES (LEFT) WITH CLOSE-UP OF ONE FARM (RIGHT). THE ALGORITHM USED FOR CLASSIFICATION WAS MAXIMUM LIKELIHOOD. SUPERIMPOSED ARE LOCATIONS OF NESTED PADDOCKS (WHITE) AND SITES (YELLOW). THE STUDY AREA IS LOCATED IN THE UPPER LACHLAN CATCHMENT OF NEW SOUTH WALES, AUSTRALIA (TOP RIGHT). CONTAINS MODIFIED COPERNICUS SENTINEL DATA (EUROPEAN SPACE AGENCY, 2023) AND THE WORLD LIGHT GRAY BASE MAP (ESRI ET AL., 2011). CREATED WITH ARCGIS PRO (ESRI, 2023). 32

FIGURE S7. CLASSIFIED MAP OF TREE COVER IN 2023 (LEFT) WITH CLOSE-UP OF ONE FARM (RIGHT). THE ALGORITHM USED FOR CLASSIFICATION WAS MAXIMUM LIKELIHOOD. SUPERIMPOSED ARE LOCATIONS OF NESTED PADDOCKS (WHITE) AND SITES (YELLOW). THE STUDY AREA IS LOCATED IN THE UPPER LACHLAN CATCHMENT OF NEW SOUTH WALES, AUSTRALIA (TOP RIGHT). CONTAINS MODIFIED COPERNICUS SENTINEL DATA (EUROPEAN SPACE AGENCY, 2023) AND THE WORLD LIGHT GRAY BASE MAP (ESRI ET AL., 2011). CREATED WITH ARCGIS PRO (ESRI, 2023). 33

FIGURE S8. CLASSIFIED MAP OF TREE COVER IN 2005 (LEFT) WITH CLOSE-UP OF ONE FARM (RIGHT). SUPERIMPOSED ARE LOCATIONS OF NESTED PADDOCKS (WHITE) AND SITES (YELLOW). THE STUDY AREA IS LOCATED IN THE UPPER LACHLAN CATCHMENT OF NEW SOUTH WALES, AUSTRALIA (TOP RIGHT). CONTAINS TREE COVER DATA FROM FISCHER ET AL. (2009) AND THE WORLD LIGHT GRAY BASE MAP (ESRI ET AL., 2011). CREATED WITH ARCGIS PRO (ESRI, 2023). 34

FIGURE S9. TREE COVER CHANGE [PERCENTAGE POINTS] ON PADDOCKS FROM 2005-2023 (LEFT) WITH CLOSE-UP OF ONE FARM (RIGHT). THE COLOURS INDICATE DECREASING TREE COVER (PURPLE), INCREASING TREE COVER (GREEN) AND APPROXIMATELY NO CHANGE IN TREE COVER (WHITE). THE STUDY AREA IS LOCATED IN THE UPPER LACHLAN CATCHMENT OF NEW SOUTH WALES, AUSTRALIA (TOP RIGHT). CONTAINS MODIFIED COPERNICUS SENTINEL DATA (EUROPEAN SPACE AGENCY, 2023), TREE COVER DATA FROM FISCHER ET AL. (2009) AND THE WORLD LIGHT GRAY BASE MAP (ESRI ET AL., 2011). CREATED WITH ARCGIS PRO (ESRI, 2023). 35

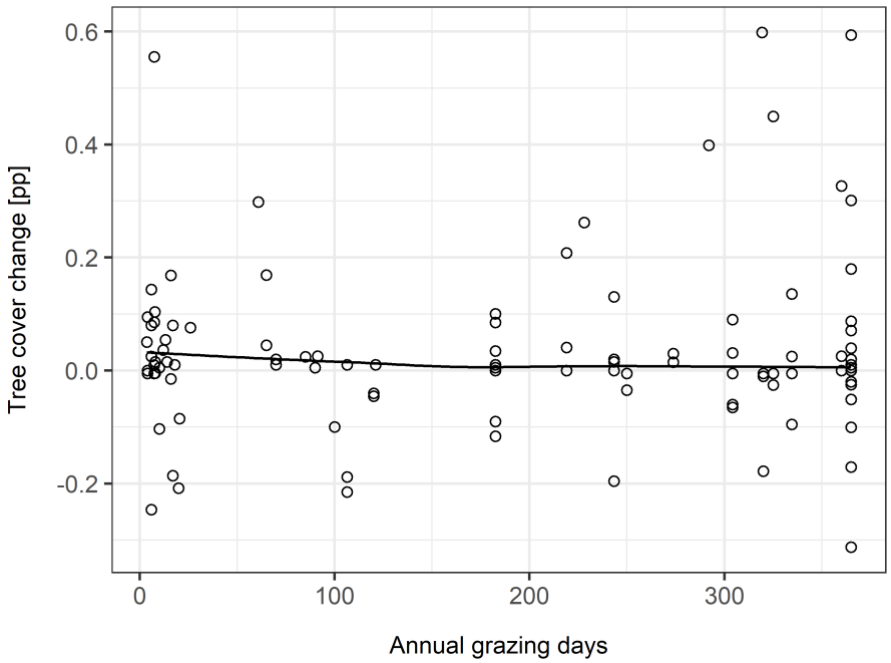


Figure S1. Scatterplot (with loess-curve) of tree cover change [percentage points] in relation to annual grazing days on sites.

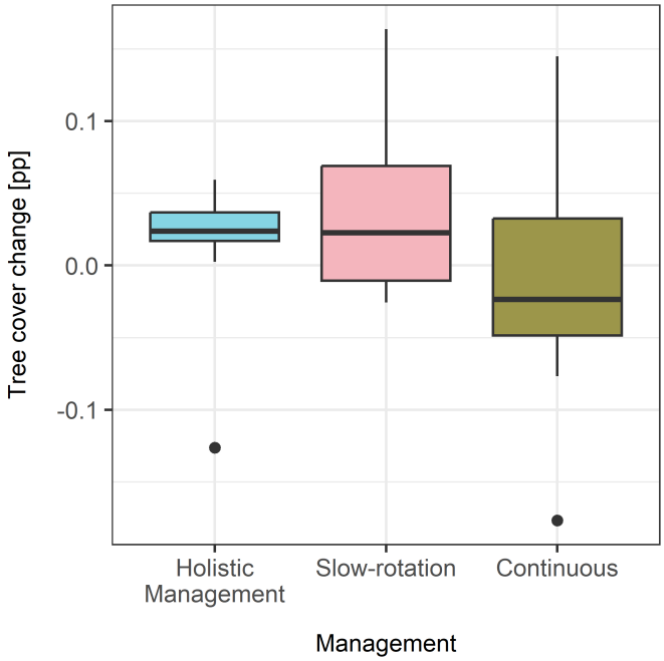


Figure S2. Boxplots of tree cover change [percentage points] on farms with different management types.

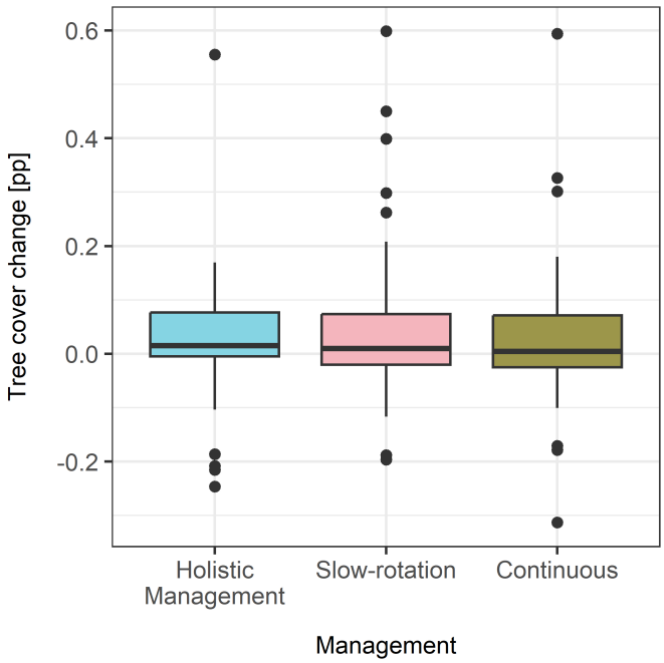


Figure S3. Boxplots of tree cover change [percentage points] on sites belonging to farms with different management types.

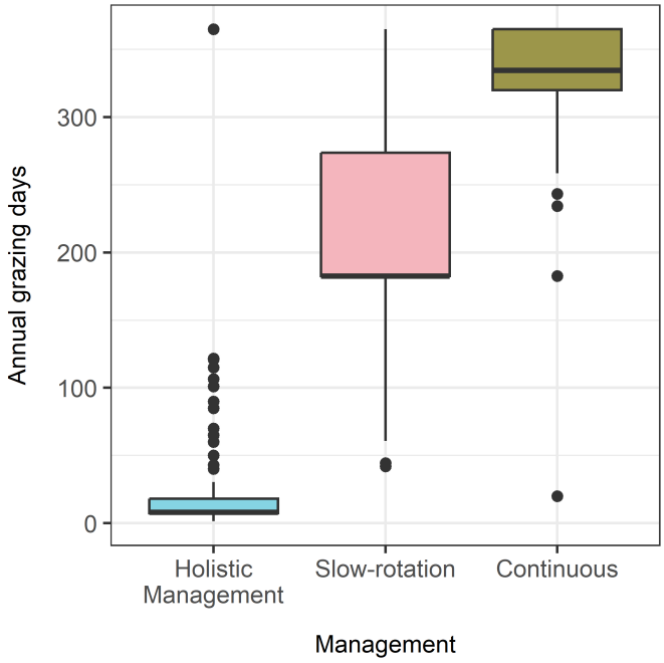


Figure S4. Boxplots of annual grazing days on paddocks belonging to farms with different management types.



Figure S5. Mosaic of Sentinel-2 images used for tree cover mapping in 2023 (left) with close-up of one farm (right). Superimposed are locations of nested paddocks (white) and sites (yellow). The study area is located in the Upper Lachlan Catchment of New South Wales, Australia (top right). Contains modified Copernicus Sentinel data (European Space Agency, 2023), the Hybrid Reference Layer (Esri et al., 2017) and the World Light Gray Base map (Esri et al., 2011). Created with ArcGIS Pro (Esri, 2023).

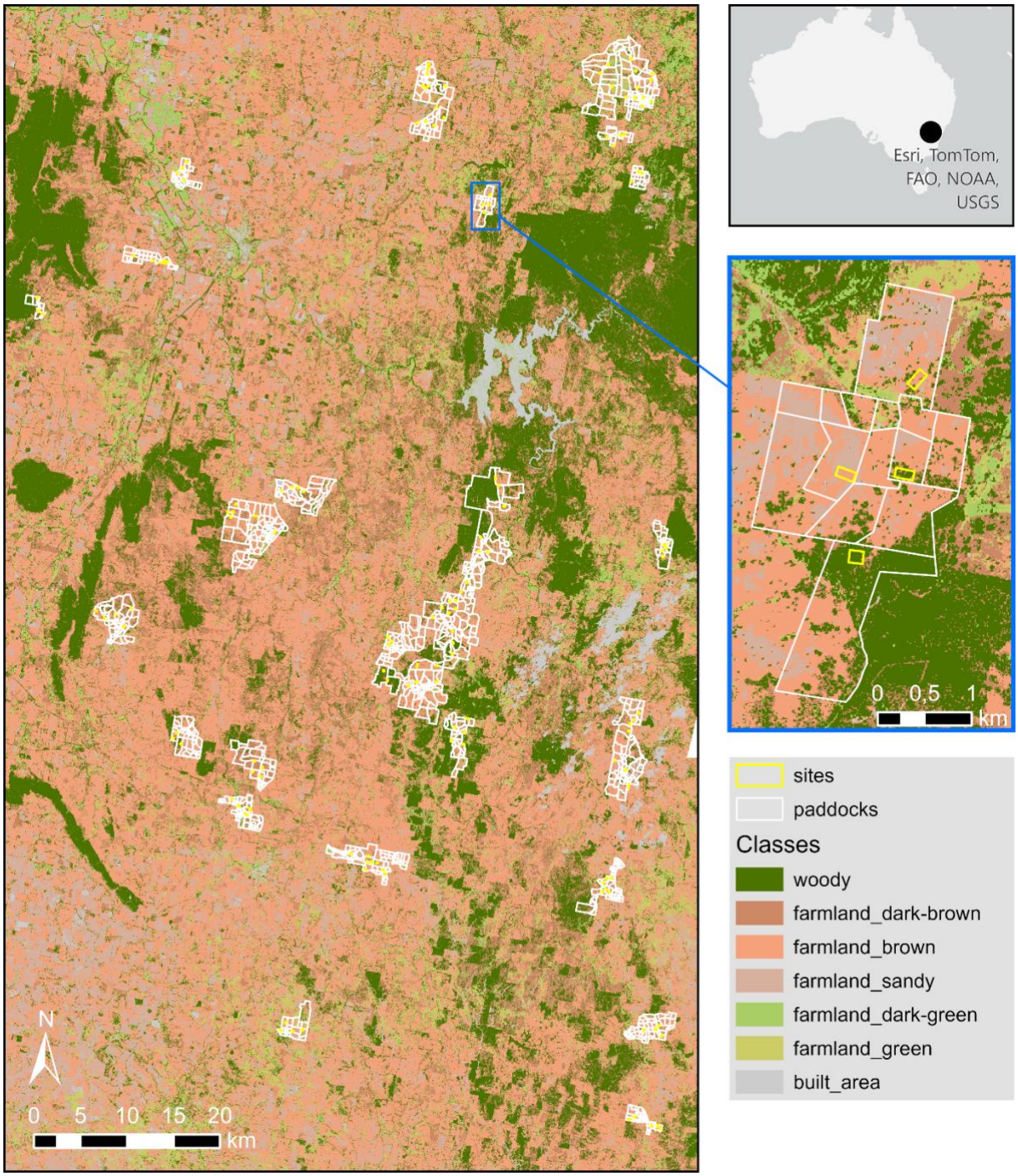


Figure S6. Classified map of land cover in 2023 using a woody and 6 non-woody classes (left) with close-up of one farm (right). The algorithm used for classification was Maximum Likelihood. Superimposed are locations of nested paddocks (white) and sites (yellow). The study area is located in the Upper Lachlan Catchment of New South Wales, Australia (top right). Contains modified Copernicus Sentinel data (European Space Agency, 2023) and the World Light Gray Base map (Esri et al., 2011). Created with ArcGIS Pro (Esri, 2023).

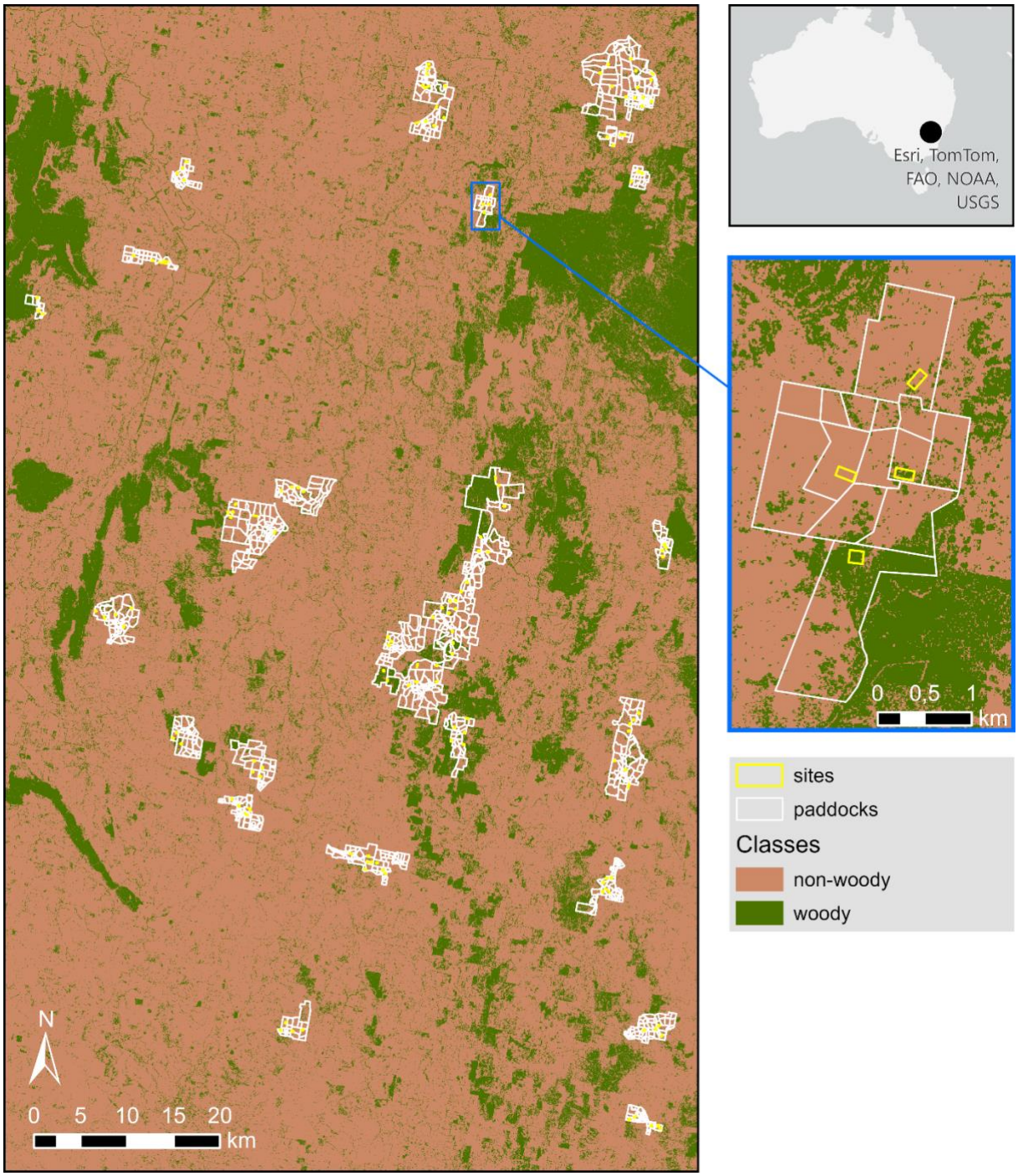


Figure S7. Classified map of tree cover in 2023 (left) with close-up of one farm (right). The algorithm used for classification was Maximum Likelihood. Superimposed are locations of nested paddocks (white) and sites (yellow). The study area is located in the Upper Lachlan Catchment of New South Wales, Australia (top right). Contains modified Copernicus Sentinel data (European Space Agency, 2023) and the World Light Gray Base map (Esri et al., 2011). Created with ArcGIS Pro (Esri, 2023).

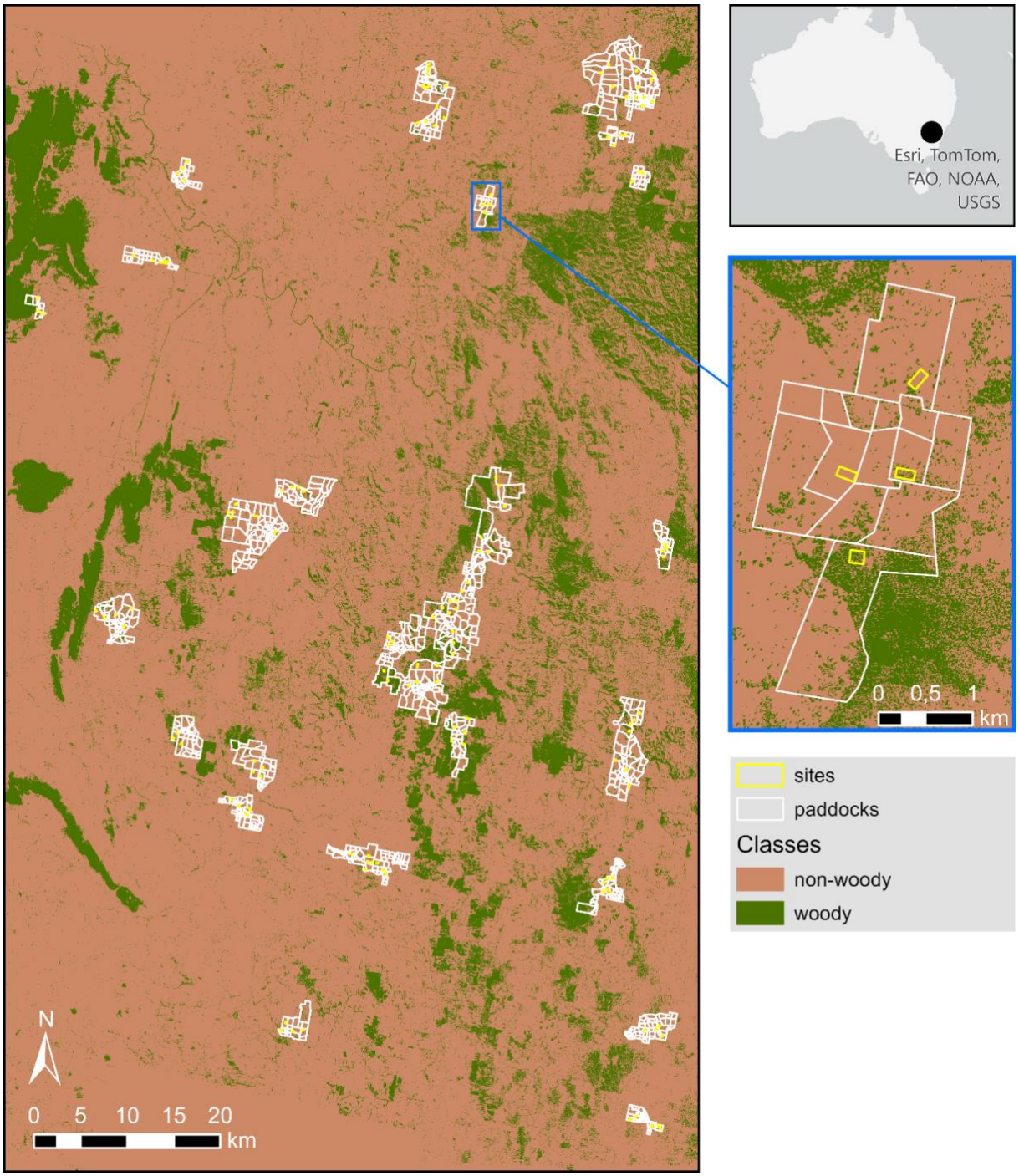


Figure S8. Classified map of tree cover in 2005 (left) with close-up of one farm (right). Superimposed are locations of nested paddocks (white) and sites (yellow). The study area is located in the Upper Lachlan Catchment of New South Wales, Australia (top right). Contains tree cover data from Fischer et al. (2009) and the World Light Gray Base map (Esri et al., 2011). Created with ArcGIS Pro (Esri, 2023).

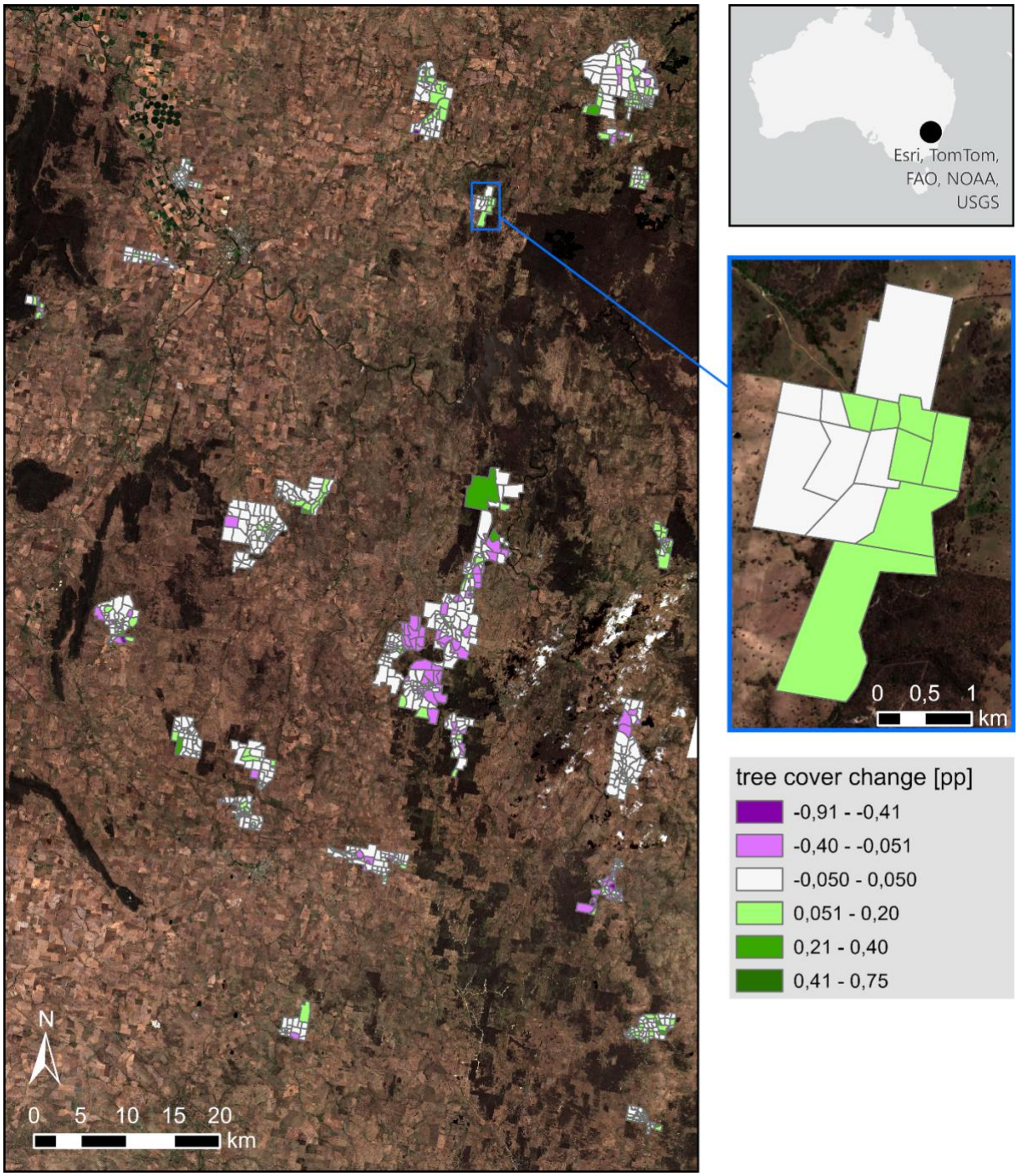


Figure S9. Tree cover change [percentage points] on paddocks from 2005-2023 (left) with close-up of one farm (right). The colours indicate decreasing tree cover (purple), increasing tree cover (green) and approximately no change in tree cover (white). The study area is located in the Upper Lachlan Catchment of New South Wales, Australia (top right). Contains modified Copernicus Sentinel data (European Space Agency, 2023), tree cover data from Fischer et al. (2009) and the World Light Gray Base map (Esri et al., 2011). Created with ArcGIS Pro (Esri, 2023).